- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

16 A particle of mass 2 kg slides down a plane inclined at 20° to the horizontal. The particle has an initial velocity of $1.4\,\mathrm{m\,s^{-1}}$ down the plane. Two models for the particle's motion are proposed.

In model A the plane is taken to be smooth.

(a) Calculate the time that model A predicts for the particle to slide the first 0.7 m. **[5]**

(b) Explain why model A is likely to underestimate the time taken. **[1]**

In model B the plane is taken to be rough, with a constant coefficient of friction between the particle and the plane.

(c) Calculate the acceleration of the particle predicted by model B given that it takes 0.8 s to slide the first 0.7 m. **[2]**

(d) Find the coefficient of friction predicted by model B, giving your answer correct to 3 significant figures. **[6]**